

Spectro Multi-Fibres

Analysis of 80-fiber scattering spectra — user guide

This application turns raw detector images into calibrated spectra, then into angular maps of the scattered light on the collection sphere. It runs in your web browser, on your own machine. No programming is needed.

Who it is for	Anyone analysing a shot campaign
What you need	Detector images, one HgAr image, an Excel shotbook, and the files describing where the fibers sit
How long	About ten minutes of setup per campaign, then the extraction runs on its own

How to read this guide. Sections 1 to 3 get you running. Sections 4 to 12 follow the nine pages of the application in the order you use them; each one tells you what to click and what you should see afterwards. Sections 13 to 16 cover changing campaign, fixing problems, and the conventions the application uses. Read 1 to 7 once, then come back to the rest as needed.

Throughout, **bold** marks something you click or type, and `monospace` marks a file name or a value.

1. Installing and starting

The application is a local web server. Nothing is uploaded anywhere: your data stays on your machine, and the browser is only used to display the interface.

Requirements

Python 3.10 or later, and these packages:

```
pip install dash dash-bootstrap-components plotly numpy scipy \
pillow tifffile openpyxl matplotlib
```

Starting

From the folder containing `run.py`:

```
python run.py
```

What you should see. A terminal window printing `Spectro Multi-Fibres -- serveur local`, and a browser tab opening on `http://127.0.0.1:8050`. If the browser does not open by itself, type that address in it.

Closing the terminal window stops the application.

→ Keep the terminal window visible while you work. If something goes wrong, the full error message appears there, and it is the first thing to look at.

The nine pages

The left-hand menu follows the order of the work. Each page states what it needs, and offers a button to the next step once it is satisfied.

Page	What it is for	Needed?
1. Input data	Point to your files; the application checks them	Always
2. Calibration	Fiber positions, tilt, pixel — wavelength	Always
3. Absolute calibration	Convert ADU into spectral energy (J/nm)	Optional
4. Extraction	Turn every image into 80 spectra	Always
5. Exploration	Look at one image, one spectrum, one fiber	As needed
6. Groups and comparisons	Compare conditions, find aberrant shots	As needed
7. Angular maps and 3D	Where the light goes on the collection sphere	As needed
8. Laser correlations	Spectral area vs energy, power, intensity	As needed
9. Exports and history	Files produced, trace of your analyses	As needed

2. The files you need

Gather these before you start. The application never guesses: if something is missing it says so rather than carrying on with a default.

File	What it is	Where it goes
shotNNN.tif	One detector image per shot, 16-bit greyscale, all in one folder	Science images folder
HgAr image	A mercury-argon lamp image, acquired in the same conditions as the shots (same binning, same sensor)	Detected in the folder, or given by hand
Final.xlsx	The shotbook: laser energies and campaign metadata, one row per shot	Detected in the folder, or given by hand
Angle reference file	Gives ϕ for each arm and θ for each port. The same file serves every campaign	Page 1, angular configuration
Fiber map(s)	For each fiber: its quadrant, its arm, its port. One file per configuration	Page 1, angular configuration
Pulse CSVs	Optional. Temporal pulse profiles, for peak power and peak intensity	Detected in the folder

A warning about image format

Every science image must be a **16-bit greyscale TIFF**, exactly as the detector wrote it. An image re-exported in colour or in 8 bits looks identical in a file browser and has the same dimensions, but its values are capped at 255 instead of 65535 and its array has the wrong shape. The application detects this and refuses the file, naming it, rather than producing a spectrum that is silently about 250 times too weak.

→ If page 1 reports images in RGBA or 8-bit format, re-export those shots from the original detector files. Until you do, they are excluded from every average, and the application says so each time.

3. Page 1 — Input data

This page tells the application where everything is, and checks that it holds together. Nothing else works until it is green.

Step by step

1. In **Folder path (copy-paste from your file explorer)**, paste the full path of the folder containing the `shotNNN.tif` images.
2. Click **Scan the folder**.
3. Read what comes back, then fill in anything it could not find.
4. Fill in the angular configuration (below).
5. Click **Validate and save the configuration**.

What you should see after Scan. A summary listing the number of science images found, the candidate HgAr images, the candidate Excel files, and any pulse CSV folder. The HgAr and Excel dropdowns are filled with what it found.

If a file was not detected, type its full path in **...or manual path to the HgAr image** or **...or manual path to the Excel file**. A manual path always wins over the dropdown.

The **Workspace folder** holds the cache, the exports and the history. Leave it empty to use the application's own workspace folder.

Angular configuration

Two kinds of file, and the distinction matters.

The angle reference file gives ϕ for each arm and θ for each port. Its layout does not change between campaigns, so it is the **only** source of angles. Put its path in **Angle reference file (.xlsx) — required**.

The campaign fiber map(s) say, for each fiber, only its quadrant, its arm and its port. Put one path **per line** in **Campaign fiber map(s) (.xlsx)**: a campaign with three configurations gets three lines, and each becomes its own angular configuration named after its file.

Then click **Read the files and build the configuration**.

What you should see. One block per file, each with a green badge giving the configuration name, the number of fibers resolved (80/80 if all went well), and the ϕ and θ ranges. Below each badge, the import report lists the sheet and columns it used, the arms it recognised with their angles, and the ports it found.

Any ϕ , θ , x, y or z columns present in a campaign file are **ignored**. They have been wrong before — one campaign wrote θ as 180 minus θ , which turns the sphere upside down. The report tells you by how much they differed, so that you know it happened rather than discovering it in a figure.

Columns are recognised by name and, failing that, by content: a quadrant column contains alpha or delta, an arm is a single letter, a port a small integer. A future campaign can rename or move its columns with nothing to change here. If the automatic reading ever picks the wrong column, the report says which ones it used, and **Manual override** lets you set them by number (1 = first column of the sheet).

Fiber numbering

One setting deserves attention because it changes the physics: **Fiber numbering in the campaign file**, that is, whether the mounting sends the fibers back reversed.

- *Not reversed* — the file's fiber 1 is the first detector trace.
- *Reversed* — the file's fiber 1 is the **last** detector trace.

This is a property of the setup, not of the file. The November 2025 campaign is reversed; the May 2026 one is not. It can also be flipped live on the 3D view, which is the easiest way to check it.

→ The test is physical: on the 3D view, the strongest signal must land near backscatter. If the intensity gradient runs the wrong way along each arm, this is the setting to change.

Validate

Click **Validate and save the configuration**. The report uses three levels, and the distinction is deliberate.

Level	Meaning
Red	Blocking. Folder missing, HgAr absent, dimensions that do not match. No button to the next step
Orange	Warning. Unusable image formats, angles in a campaign file that disagree with the reference. You can carry on
Plain	Information: what was found and loaded

What you should see when it works. A green or orange panel ending with the button **Next step: Calibration**. If that button is missing, something is red.

4. Page 2 — Calibration

This page establishes where the fibers sit on the detector, how much the image is tilted, and how a pixel column maps to a wavelength. Everything downstream rests on it.

The default parameters are the validated ones. Change them only with a reason.

Step by step

1. Check **Number of fibers** (80) and **Extraction half-width (px)**.
2. Leave λ **calibration method** on *Automatic (comb-matching, recommended)* unless you have a reason not to.
3. Click **Run calibration**.
4. Read the detected HgAr lines and the fit quality.
5. If neutral-density filters were used, go to **ND filter correction** and click **Detect the ND values in the shotbook**, then **Save & check the ND files**.
6. In **Automatic fiber detection**, click **Show image + fibers** and compare the detected positions with the image before accepting them.

If the images are not straight

The extraction expects the fiber traces to run **horizontally**. Depending on how the detector was mounted, they may come out vertical, and the whole campaign would then be unreadable.

Two dropdowns fix this: **Rotate science images** and **Rotate calibration image**, each offering *None*, *90° clockwise*, *180°* and *90° counter-clockwise*. They are independent, because the lamp and the shots are not always acquired in the same orientation.

Look at the **Detector view** tab of page 5, or at **Show image + fibers** on this page: if the traces run top to bottom instead of left to right, rotate.

→ Worked example: in the May 2026 campaign, the science images must be rotated **90° counter-clockwise**, while the HgAr image needs no rotation.

Set the rotation **before** running the calibration. Changing it afterwards invalidates the fiber positions and the wavelength axis, and both must be recomputed.

What you should see after Run calibration. The **Auto-detected HgAr lines** card fills with the reference lines it matched and a **Calibration RMS**. A small RMS and a sensible list of lines mean the pixel-to-nanometre conversion can be trusted. If the RMS is large or lines are missing, the lamp image is probably not from the same optical configuration as the shots.

What you should see after Show image + fibers. The detector image with the 80 detected traces drawn on top. Every trace should sit on a real line of signal. If some are off, use **Re-detect fibers**, or correct them and click **Apply corrections**.

→ If the HgAr image was acquired with a different binning from the shots, the calibration cannot apply. Page 1 refuses this case rather than producing a wavelength axis that does not match the data.

5. Page 3 — Absolute calibration (optional)

This page converts the ADU spectra into physical spectral energy, in joules per nanometre, per fiber and per wavelength. It needs a calibrated lamp and a power-meter measurement. **Skip it if you do not have them** — everything else works in ADU.

Step by step

The four cards are numbered in the interface (1 · to 4 ·); follow them in order.

1. **Measurement files** — give the after-fiber lamp image (.tiff), the before-fiber lamp spectrum (.spf2 or a two-column .csv), the per-fiber power-meter readings (.txt, lines fiber / value), and optionally a dark image taken at the same exposure with the lamp off.
2. **Power meter and geometry** — say what the meter measured, in which unit, and how the geometry is entered (distance, aperture).
3. **Optics on the lamp image** — the transmission of the elements sitting between the lamp and the fibers.
4. **Smoothing and bad fibers** — choose the smoothing method and list the fibers to replace by the mean curve.
5. Click **Build the absolute calibration**.
6. In **Apply to a science shot (check)**, pick a shot and click **Convert to J/nm**.

What you should see. The **Calibration curves** card draws $J \times nm^{-1} \times count^{-1}$ against wavelength, one curve per fiber, and the **Calibration map** shows them all at once. Curves should be smooth and of similar shape; a fiber whose curve jumps out is a candidate for the bad-fiber list.

Then read the **Numeric audit of the chain**. It walks through each factor of the calculation and its order of magnitude. This is how you catch a factor of a thousand before it reaches a figure — do not skip it.

→ If you skip this page, the J/nm option stays greyed out everywhere and only ADU is offered. This is deliberate: a calibration belongs to the campaign that produced it, and the application will not offer physical units it cannot justify.

6. Page 4 — Extraction

Turns every detector image into 80 calibrated spectra. This is the long step, and it runs in the background: you can leave the page and come back.

Step by step

1. Read **Status**. It says what is ready and what is missing.
2. In **Start a run**, choose **Images to process** — a single image to test your settings, or the whole campaign. You can restrict the range with **From shot #** and **to shot #**.
3. Click **Start**.
4. Watch **Progress**. **Stop** interrupts a run; what is already extracted stays in the cache.

What you should see. A progress count that advances, then a summary. Any shot that could not be read is listed with the reason — most often an image format problem, named explicitly.

→ Test on one image first. If the traces or the wavelength axis are wrong, you would rather find out after one image than after two hundred.

The cache, and how to reuse it

Extracted spectra are written to a cache folder so that they are computed once. The folder is keyed on the **content** of the HgAr image and on the calibration parameters, which means copying files or moving the campaign folder no longer orphans a perfectly good cache.

Cache management lists every cache it can see, with where it came from: which campaign, which HgAr image, when it was built. A cache from another campaign is labelled *other campaign*. Each non-empty cache carries a **Use this one** button, which makes the extraction skip the shots it already contains.

The other buttons — **Clear the current cache**, **Delete older caches**, **Delete everything** — do exactly what they say and cannot be undone.

→ A cache built before this feature existed shows *no provenance recorded*. It is still usable: press **Use this one**.

7. Page 5 — Exploration

Free inspection of one image at a time. Choose an **Image** at the top; the tabs then show it from different angles.

Tab	What it shows	Use it to
Detector view	The raw image	Check the traces and the background
All 80 spectra	A heat map, all fibers at once	Spot a dead or saturated fiber
Fiber inspection	One fiber's spectrum	Look at a line, a shape, a level
Fiber comparison	Several fibers overlaid, optionally normalised	Compare neighbours
Temporal evolution	The signal across shots	See a drift over the campaign
SNR map	Signal-to-noise per fiber and wavelength	Decide which fibers to trust

For **Fiber inspection**, type a number in **Fiber (1–80, detector order)**. For **Fiber comparison**, list several: 38, 39, 40, 41. For the **SNR map**, click **Compute the SNR map**.

Images whose format prevents extraction stay in the list but are greyed out and labelled unusable file. They are shown rather than hidden so that you can see they were provided; selecting one gives an explicit red message naming the problem, instead of leaving the previous figure on screen.

Units

The ADU / J—nm switch is global: it follows you from page to page. It is available only if the absolute calibration has been done for the current campaign.

8. Page 6 — Groups and comparisons

A group is a set of shots taken under the same experimental condition. This page compares groups and, just as importantly, tells you when a shot does not belong.

Building groups

The quickest route is automatic. In **Automatic groups from Final.xlsx**, set **Central 2 ω energy**, a **Tolerance ($\pm$ %)** and a **Min. group size**, then click **Analyse**. You get a proposal; accept it with **Replace all groups with this proposal**, or with **Add to the existing groups** to keep what you already had.

To build one by hand: **New group**, give it a **Name** and a **Colour**, list its shots in **Shots of the group** (133-138 or 133,134,137), then **Save this group**. **Delete this group** removes one; **Restore the campaign groups** goes back to the automatic proposal.

Comparing

1. Select the groups in **Groups to compare**.
2. Choose the **Fiber (1–80)** and the λ **range (nm)**.
3. Choose **Spread** — standard deviation, or standard error of the mean.
4. Set the **Outlier tolerance ($\pm$ % around the group median)** if the default of 40 % does not suit.
5. Click **Compare**.

What you should see, in this order: the outlier check, the individual spectra, the mean with its band, a 3D surface over fibers and wavelengths, and a table of the integral of each group.

The outlier check

It is always displayed, whether or not anything is found, and it comes first so that you read it before the curves.

It works on **two** measurements per shot: the integral of the displayed fiber, and the integral summed over the 80 fibers. The second is what catches a shot that is weak overall when the displayed fiber happens to be noisy.

A shot is named as aberrant when either measurement departs from its group median by more than the outlier tolerance, or by more than 4σ on a robust estimate when the group is otherwise very tight. The red banner names the shots and says why — for instance *$\times 0.35$ the group median on the total over the 80 fibers*. You can then decide whether to edit the group and drop the aberrant shots — select it in **Group to edit**, remove them from **Shots of the group**, click **Save this group**, then **Compare** again.

Below the banner, a table gives every shot's values and its ratio to the median, even when nothing is flagged. This is deliberate: *no alert* should be something you can check, not a silence you have to trust.

→ A shot excluded because its file is unusable is reported separately, above everything else. It is absent from the averages, and you should know that before reading them.

The Si map

Spectral-integral map: Si % $\times$ pulse profile crosses the silicon percentage with the pulse profile. Set the two axes, then click **Compute the map**.

9. Page 7 — Angular maps and 3D view

Where the scattered light goes on the collection sphere. Three tabs: **2D map** (θ , ϕ), **Interactive 3D sphere**, and **Centroid vs θ (fixed ϕ) + export**.

Step by step

1. Choose the **Image**.
2. Choose the **Angular configuration** — it can be resolved automatically from the shotbook, or picked by hand.
3. Set the λ **range (nm)** and the **Metric**.
4. Adjust **RBF smoothing** and **Coverage (°)** if the surface looks too smooth or extends too far from real fibers.
5. Click **Show**.

The 3D view is mouse-controlled: drag to rotate, scroll to zoom. The exported image, from the camera icon at the top right of the figure, follows the orientation currently on screen.

Fibers to exclude, 1–80 removes fibers from both the points and the interpolation. Use it, for instance, to see the side-scatter without the backscatter saturating the colour scale.

Mounting conventions

Two settings describe the setup rather than the data, and both are populated from what the configuration file actually contains.

Negate θ / ϕ for the quadrant(s) — the menus offer only the quadrants really present in the chosen configuration, and open on the rule that configuration was imported with. If a menu is empty, that configuration carries no quadrant information.

Fiber order — reversed or not, as described in section 3. Flip it here to see the effect immediately.

Under the figure, a line always states which conventions produced what you are looking at: the geometry used, the fiber order, the sign rule. Nothing is applied silently.

→ The plot axes are **not** the laboratory axes. In the pipeline conversion, the backscatter direction is the figure's +y, and the angle to it is exactly θ . The green arrow shows the beam; it is imposed by the experiment and is not a setting.

What a correct result looks like. The strongest signal sits near backscatter, and the intensity falls away as you move towards side-scatter. If it rises instead, check **Fiber order** first, then the sign rule.

10. Page 8 — Laser correlations

Correlates the spectral area with the laser drive: 2ω energy, peak power, or peak intensity. Energy comes from the Excel shotbook.

Step by step

1. In **Parameters**, choose the **X axis**, the **Shot source** (groups, or an explicit list of shots), the **Fiber** (empty = mean of all 80) and the λ **range**.
2. Choose the **Fit type**, its **Degree** if it is a polynomial, and the **Confidence** level.
3. For peak intensity, fill in **Focal spot**: the super-Gaussian **Order n** and **Radius w (μm)**.
4. For peak power, load the pulse profiles: set the **CSV header lines** and the rising and falling edge thresholds, then click **Show**.
5. Click **Compute**.

What you should see. The scatter plot with the fitted law and its confidence band, and a **Fit details** card giving the parameters. Read those parameters before believing the curve.

Looking at the pulses themselves

The **Pulse profiles** card is not only a prerequisite for peak power: it is also how you check that the pulses are what you think they are. Choose a **Mode**, then click **Show**.

Mode	What you give	What it is for
Single pulse	A Shot number	Inspect one profile in detail
All pulses of a category	A Category	Check the consistency of a whole class of shots
All pulses of a saved group	A Group defined on page 6	Check a group before you use it in a comparison

Overlaying a group is the quick way to spot a pulse that does not belong: a different duration, a double peak, a truncated rising edge. A shot whose pulse is aberrant will drag the peak power and the peak intensity with it, so it is worth removing from the group on page 6 before computing the correlation.

→ The rising and falling edge thresholds, expressed as a fraction of the peak, define where the pulse is considered to start and stop. If the duration looks wrong for every shot at once, it is usually these thresholds, not the data.

→ Peak power and peak intensity need the pulse CSVs and the focal spot. Without them, only the energy correlation is available.

11. Page 9 — Exports and history

- **Produced files** — everything written during the session, ready to download. **Refresh the list** updates it.
- **'Publication' figures** — the original matplotlib rendering, for figures going into a paper. Choose the **Figure**, the **Image** and the **Fiber** if they apply, then **Generate the PNGs**.
- **Graph display preferences** — **Font size** and **Theme** for the interactive figures. **Apply** to activate.
- **Analysis history** — the trace of what you did, which is what makes a result reproducible three months later.

Interactive graphs also export directly from their own toolbar: the camera icon in the top-right corner of each figure.

12. Changing campaign

Everything computed — wavelength calibration, absolute calibration, energy table, metadata, angular configurations — belongs to one campaign, identified by its images folder, its HgAr image and its shotbook.

When you validate page 1 with different paths, the application detects the change, clears what belonged to the previous campaign, and lists in orange what it cleared. Your paths and preferences are kept, and you do not need to restart the application.

→ Redo the calibration (page 2) before anything else. The extraction cache of the other campaign is kept on disk and will be reused if you come back to it.

13. When something looks wrong

Symptom	What it usually is
A page does not open	Look at the terminal window: the full error is there. A stale <code>__pycache__</code> after an update gives the same symptom — delete those folders and restart
No Next step button on page 1	Something is red, not orange. Only red is blocking
J/nm is greyed out	The absolute calibration has not been done for <i>this</i> campaign. That is the intended behaviour
The extraction re-does everything	The cache key changed. Go to Cache management and press Use this one on the populated cache
A shot is far weaker than the rest	Check its file format first. An 8-bit or colour re-export is capped at 255 instead of 65535
The 3D gradient runs the wrong way	The fiber order. Flip Fiber order on page 7: the strongest signal must land near backscatter
One quadrant sits apart from the rest	The sign rule. Check Negate θ against the quadrants the configuration actually contains
A quadrant menu is empty	That configuration carries no quadrant information — re-import it from page 1
A spectrum looks like the previous one	An unusable image now returns an explicit red message instead of leaving the previous figure. If you see this, you are on an older build

14. Conventions, for reference

Angles

Every campaign uses the same reference file and the same conversion. Only two things vary, and both are explicit settings.

Fixed or setting?	What
Fixed	The angle reference file: ϕ per arm, θ per port
Fixed	The conversion from angles to x, y, z. It is the pipeline function, unchanged, and it is not selectable
Fixed	The beam direction, imposed by the experiment. Backscatter is the figure's +y, and the angle to it is exactly θ
Setting	Fiber order — reversed or not, per configuration
Setting	Negative angle per quadrant, per configuration

Units

ADU are raw detector counts. **J/nm** is spectral energy, available only after the absolute calibration of the current campaign. In J/nm a typical peak reads around 70 p , that is $7 \times 10^{-11} \text{ J/nm}$. If you ever see a value in μ where you expect p, a factor of a million has crept in somewhere — report it.

Fiber numbering on the figures

The number written next to each point is the **extraction index**: the order of the traces on the detector, starting at 1. It is not necessarily the fiber number in your configuration file — that is exactly what the fiber order setting is about.

What the application will not do

It does not guess. When a piece of information is missing, or two files disagree, it says so and names what it found rather than filling in a plausible default. Warnings that do not block are orange; only genuinely blocking problems are red. If you ever find it choosing silently on your behalf, that is a bug worth reporting.

15. Editing this guide

This document is written in Markdown, in docs/UserGuide.md. Edit that file with any text editor, then rebuild the PDF:

```
python docs/md2pdf.py docs/UserGuide.md SpectroMultiFibres_UserGuide.pdf
```

The Markdown subset understood by the converter is deliberately small: headings with #, paragraphs, - bullet lists, 1. numbered lists, > call-outs, | tables, fenced code blocks, **bold**, *italic*, ``` code and --- for a page break. A few Greek letters and symbols are written as θ , ϕ , μ , σ , ω , $\times$, $^\circ$ and --- .

16. Getting help

When reporting a problem, include what you did, what you expected, what happened, and the contents of the terminal window. For anything angular, add the import report from page 1 and the convention line printed under the 3D figure — between them they say exactly which conventions were used.